EFU-Phosphorus Academic & Standard Use Licence
(EFU-P-ASUL v1.0 – Revised, Research Framework Version)

Applies to: EFU-Phosphorus Standard v1.0 and subsequent minor revisions
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Base licence: EFU Universal Licence v1.0 (CC BY 4.0 compatible)

Status and scope

EFU-P is an open, experimental research and measurement framework and a proposed human-referenced reference unit for phosphorus flows. It is a thought-experiment-level analytical construct, not a legal standard, certification scheme or regulatory requirement, and it shall not be interpreted or presented as such.

All EFU-P indicators, baselines and thresholds are **provisional research parameters**. They are intended for exploratory analysis, methodological comparison and academic or policy discussion, and are subject to revision, correction and empirical calibration as new data and methods become available.

This licence governs the use of the EFU-P term and reference definition in research and applied contexts, while remaining fully compatible with the freedoms and obligations of CC BY 4.0.

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1. Grant of rights

This licence grants open, worldwide, royalty-free permission to:

- use, reproduce and distribute the EFU-Phosphorus research framework and reference definition (EFU-P);
- apply EFU-P in academic research, education, policy analysis, environmental management, agriculture and wastewater studies;
- include EFU-P in peer-reviewed publications, reports, theses, datasets and decision-support tools;

provided that proper attribution is given (Section 2) and the core research definition is respected and transparently handled (Section 3).

Translation clause. The text of this licence may be translated into other languages, provided that any translation is made available under the same terms and clearly indicates that the English version is the reference text in case of divergence.

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2. Attribution requirement

Any use of EFU-P must include attribution in the following form (or an equivalent scholarly citation):

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Attribution may be placed in any reasonable manner (footnote, bibliography, metadata, software "About" page, etc.) and must not imply endorsement by the author or the steward organization.

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3. Core research definition and consistency

For the purposes of this research framework, the EFU-Phosphorus reference definition is set as follows:

1 EFU-P = 0.55 kg elemental phosphorus (P) per year per person,
defined as the average adult human phosphorus excretion rate.

This definition is a **research baseline**, not a legally binding or regulatory standard. It is intended to provide a consistent human-referenced normalisation constant for comparative work.

Permitted actions:

- scaling EFU-P to any system (farms, rivers, nations, industries);
- conversions from phosphorus compounds (P_2O_5, PO_4^{3-}, etc.) after conversion to elemental P;
- integration with EFU-N, EFU-C, EFU-W and other EFU research standards.

Prohibited without explicit renaming:

- redefining EFU-P with a different baseline while still calling it "EFU-P";
- using P_2O_5-based values without prior conversion to elemental P;
- using the term "EFU-P" to denote a metric that is not human-referenced.

Labelling requirement for deviations. Any modification of the core baseline must be explicitly labeled, for example:

"Modified EFU-P (regional baseline = 0.62 kg P yr^{-1} $person^{-1}$)".

Such variants must not be presented as the EFU-Phosphorus Standard v1.0, but only as **derived or modified EFU-P baselines**.

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4. Extensions, models and derivative works

Derivative works are explicitly allowed, including but not limited to:

- regional EFU-P baselines and parameter sets;
- sector-specific EFU-P modules (agriculture, mining, wastewater, food systems);
- legacy phosphorus models and long-term soil/sediment dynamics;
- coupling with nitrogen, carbon and water EFU research standards.

All derivative works must:

- clearly distinguish the original EFU-P reference definition from local extensions or modifications;
- cite the EFU-Phosphorus Standard v1.0 as the research baseline in methods, documentation or footnotes.

Derivative works may be released under additional or more specific terms, provided that the original EFU-P material remains available under EFU-P-ASUL v1.0 (and its CC BY 4.0-compatible base licence).

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5. Non-endorsement clause

Use of EFU-P under this licence does **not** imply endorsement by:

- István Simor;
- the Nookratic Bloc;
- any affiliated contributors.

No endorsement claim may be made without explicit written permission from the author or steward. This clause applies to all public communications, marketing materials, policy recommendations and similar contexts.

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6. Commercial and regulatory use

This licence permits:

- inclusion of EFU-P in government reports and regulatory impact assessments;
- use in support of environmental regulation and nutrient-management policies;
- use in consultancy studies, commercial research and analysis, as part of reports, tools or services.

This licence does **not** automatically grant:

- trademark-like control over the term “EFU-P”;
- exclusive certification schemes based on EFU-P;
- rebranding EFU-P as a closed or pay-walled metric.

Such uses require a separate written agreement with the steward organization.

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7. Relationship to CC BY 4.0

This licence is fully compatible with the Creative Commons Attribution 4.0 International (CC BY 4.0) licence and inherits its freedoms and obligations. It adds domain-specific safeguards regarding:

- protection and clarity of the EFU-P research baseline;
- unit consistency (elemental phosphorus, human-referenced baseline);
- prevention of silent or misleading redefinition of the EFU-P unit in scientific and applied work.

In case of ambiguity, this licence governs the use of the EFU-P **term and reference definition**, while CC BY 4.0 governs general copyright, attribution and reuse.

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8. Governing principle

EFU-P is intended as an **open, non-proprietary scientific construct and research reference unit**, not as a formal or mandatory standard. This licence exists to ensure that EFU-P:

- remains comparable across studies, sectors and jurisdictions;
- stays human-referenced and intuitively interpretable;
- can function as a shared, open language for phosphorus sustainability and planetary stewardship;
- remains open to future revision, empirical refinement and recalibration as the broader EFU Research Framework evolves.

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9. Disclaimer of warranty and liability

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Rights may be reinstated if the breach is cured within thirty (30) days of receiving written notice of the violation, in line with the restoration principles of CC BY 4.0.

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11. Governing law

This licence shall be governed by and construed in accordance with the laws of Switzerland, without regard to its conflict-of-law principles.

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12. Contact and permission requests

For any permission requests (e.g. trademark licensing, additional commercial arrangements) or clarification, please contact:

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